

Improving procedural and one-year outcome after contact force-guided pulmonary vein isolation: the role of interlesion distance, ablation index, and contact force variability in the ‘CLOSE’-protocol

Thomas Philips¹, Philippe Taghji¹, Milad El Haddad², Michael Wolf¹, Sébastien Knecht¹, Yves Vandekerckhove¹, René Tavernier¹, and Mattias Duytschaever^{1,2*}

¹Department of Cardiology, Sint-Jan Hospital Bruges, Ruddershove 10, 8000 Bruges, Belgium; and ²Department of Internal Medicine, Ghent University, De Pintelaan 185, 9000 Ghent, Belgium

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Aims

We have recently shown that a contact force (CF)-guided ablation protocol respecting region-specific criteria of lesion contiguity and lesion depth (‘CLOSE’ protocol) is associated with high incidence of acute durable pulmonary vein (PV) isolation (PVI) and a high single-procedure arrhythmia-free survival at 1 year. In the present study, we compared efficiency, safety, and efficacy of ‘CLOSE’-guided PVI to conventional CF-guided PVI (CONV-CF).

Methods and results

Fifty consecutive paroxysmal atrial fibrillation (AF) patients underwent PV encircling using a CF-sensing catheter targeting an interlesion distance (ILD) ≤ 6 mm and ablation index (AI) ≥ 400 and ≥ 550 at posterior and anterior wall (‘CLOSE’ group). Results were compared to the last 50 patients undergoing ‘CONV-CF’. All patients underwent adenosine testing after PVI. Arrhythmia recurrence was defined as any atrial tachyarrhythmia (ATA) >30 s on Holter at 3, 6, and 12 months. Clinical characteristics did not differ. Contact force variability was comparable in between both groups (proportion of applications with intermittent contact 2% in ‘CLOSE’ vs. 1% in CONV-CF, $P=0.67$). In the ‘CLOSE’ group, procedure time and radiofrequency (RF) time per circle were shorter (respectively 149 ± 33 min vs. 192 ± 42 min, $P<0.0001$ and 18 ± 4 min vs. 28 ± 7.5 min, $P<0.0001$) and incidence of adenosine-proof isolation was higher (97% vs. 82%, $P<0.001$). No complications were observed in the ‘CLOSE’ group, one tamponade in the ‘CONV-CF’ group. At 12 months, single-procedure freedom from ATA was 94% in ‘CLOSE’ vs. 80% in ‘CONV-CF’ group ($P<0.05$). In both groups, the majority of reconnections at repeat were associated with either ILD >6 mm and/or AI $<400/550$ (100% vs. 83%, $P=0.99$).

Conclusion

‘CLOSE’-guided PVI improves procedural and 1 year outcome in CF-guided PVI while shortening procedure time. Improvement cannot be explained by differences in CF variability and is most likely due to the strict application of criteria for contiguity and ablation index. A randomized controlled trial is needed to exclude the possible contribution of a learning curve.

Keywords

Atrial fibrillation • Contact force • Pulmonary vein isolation • Lesion depth • Contiguity • CLOSE protocol

* Corresponding author. Tel: +32 5045 2670; fax: +32 5045 2679. E-mail address: mattias.duytschaever@azsintjan.be

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What's new?

- The 'CLOSE' strategy is a novel contact force (CF)-guided approach to enclose the pulmonary veins (PVs) with contiguous and optimized radiofrequency lesions.
- Compared to a historical control group of conventional CF-guided pulmonary vein isolation, this novel approach is associated with shorter procedure time, higher incidence of acute durable isolation of the PVs and higher single-procedure arrhythmia-free survival at 1 year.
- Improvement cannot be explained by differences in CF variability and is most likely due to the strict application of criteria for contiguity and ablation index.

Introduction

Pulmonary vein isolation (PVI) is the cornerstone for successful treatment of paroxysmal atrial fibrillation (AF).^{1,2} We have recently shown that a contact force (CF)-guided ablation protocol aiming to enclose the veins with contiguous (interlesion distance ≤ 6 mm, ILD) and optimized radiofrequency (RF) lesions (ablation index $\geq 400/550$, AI) is associated with a high incidence of acute durable PV isolation and a high single-procedure arrhythmia-free survival at 1 year (the 'CLOSE' protocol).^{3,4}

Randomized data on the superiority of 'CLOSE'-guided PVI are lacking. In the present study, we provided comparative data on the efficiency, safety, and efficacy of 'CLOSE'-guided PVI to a historical control group of conventional CF-guided PVI.

Methods

Study subjects

From August 2013, all patients undergoing CF-guided ablation for AF at the St-Jan Hospital Bruges are followed in a prospective database approved by the local institutional review committee. This database implies collecting patients' written informed consent, a detailed case report form of the procedure, and follow-up with Holter-electrocardiograms (ECGs) at 3, 6, and 12 months. In this article, we present the analysis of the first 50 cases performed according to the 'CLOSE'-criteria. Results were compared to the last 50 cases performed with conventional CF-guided PVI (in which the operator was blinded to AI, and no formal ILD criterion was put forward).

'CLOSE'-guided pulmonary vein isolation

Pulmonary vein isolation was performed by four operators. In anaesthetized patients, oesophageal temperature monitoring (SensiTherm™, St Jude Medical Inc., St. Paul, MN, USA) was performed. To avoid discomfort, we did not use temperature monitoring under conscious sedation. Intravenous heparin (100 IU/kg) was administered and continuously infused (ACT > 300 s). After transeptal puncture, a decapolar Lasso catheter and a CF catheter (Thermocool SmartTouch®, Biosense-Webster Inc., Diamond Bar, CA, USA) were positioned in the left atrium, and calibration of CF and respiratory gating were performed. Then, we predefined the RF circle around the PVs (nephroid shape) on the 3D-geometry (Carto System®, Biosense Webster Inc.). Point-by-point RF delivery was performed during sinus rhythm aiming for a contiguous circle

enclosing the veins. Real-time automated display of RF applications (Visitag®, Biosense Webster Inc.) was used with predefined settings of catheter stability (3 mm for 8 s) and minimum CF (30% of time >4 gr). Radiofrequency was delivered (EP Shuttle ST-3077, Stockert GmbH, Freiburg, Germany) in a power-controlled mode (without ramping) aiming for 35 W (irrigation flow up to 30 cc/min) irrespective of posterior or anterior ablation. Reasons for occasional lowering of RF power were high CF at baseline (≥ 30 g) and increase in the tip T° during RF (>40°). Radiofrequency was delivered until an AI of ≥ 400 at the posterior wall/roof and ≥ 550 at the anterior wall (Figure 1, right panel). Once the target AI was reached, the RF application was stopped, and the catheter was moved to an adjacent spot. Only after verifying stability, CF and ILD, a new application was started (no catheter dragging). In case of instability leading to an RF tag not reaching the AI target, a new RF application reaching the AI target was applied. Maximal ILD between two neighbouring lesions was ≤ 6 mm (Figure 1, right panel). In case of chest pain or intra-oesophageal T° rise >38.5 °C during posterior wall ablation, RF delivery was stopped at an AI of 300 (but not before). In the absence of first-pass isolation (i.e. no isolation after completing the circle), touch-up ablation was delivered until PVI. After PVI, we waited for 20 min after which adenosine (dose resulting in atrio-ventricular block) was given four times (with the Lasso in its corresponding position). In case of reconnection during waiting time or during adenosine (acute PVR), the site of reconnection was allocated and treated with touch-up ablation until PVI was resistant to subsequent adenosine challenge. In case of pre-procedural or procedural documentation of typical flutter, cavotricuspid isthmus (CTI) ablation was performed until bidirectional block.

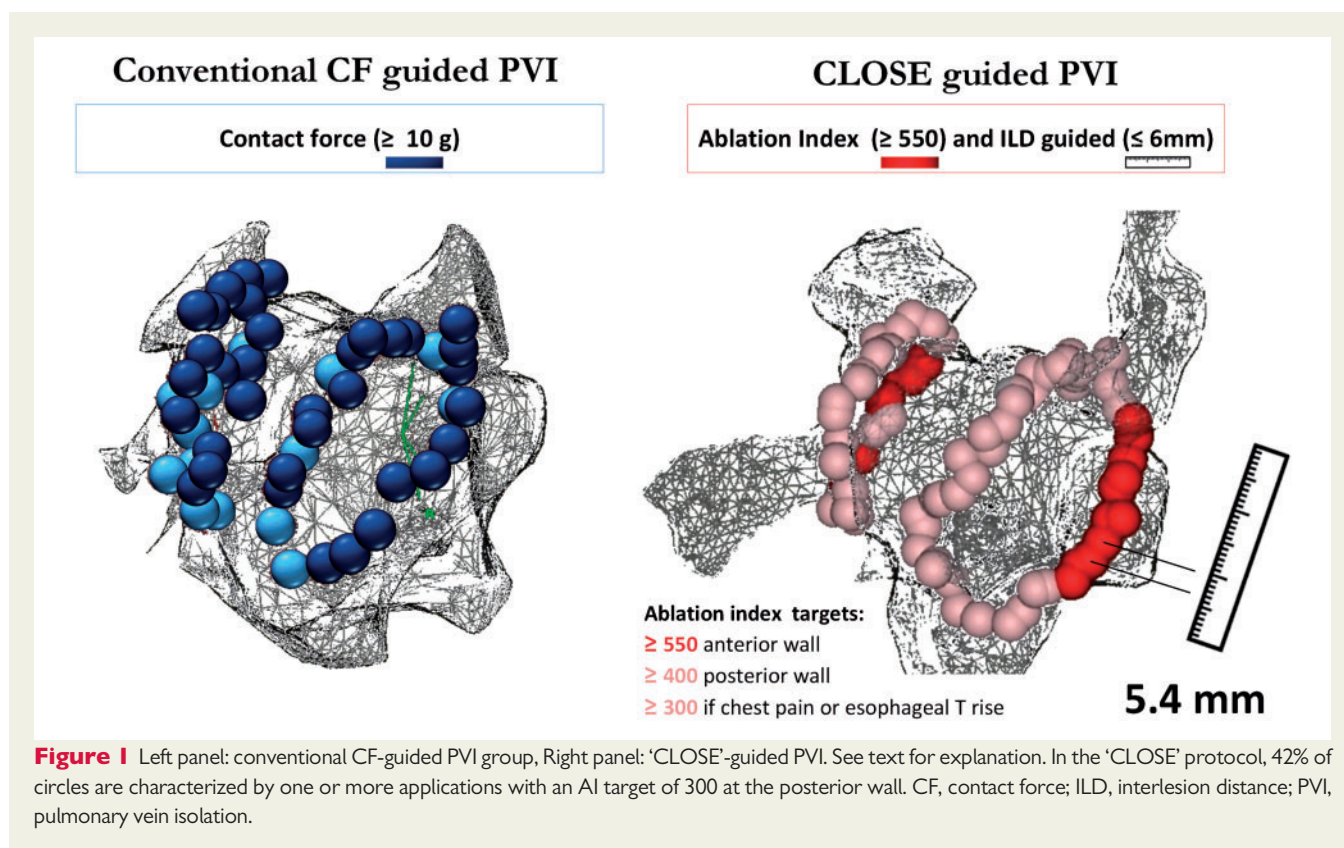
Conventional contact force-guided pulmonary vein isolation procedure

Ablation was performed by the same four operators, using the same anaesthesia protocol, transeptal access, and catheters. Also identical were calibration of CF catheter, respiratory gating, reconstructing of 3D-geometry, the predefined nephroid RF circle, Visitag® settings, RF generator, and RF mode. In contrast to 'CLOSE', RF was delivered for 30 s (at 25 W) at the posterior wall and roof, and for 60 s (at 35 W) at the anterior wall. According to data from the EFFICAS studies, CF target was >10 g (Figure 1, left panel). No specific force-time integral (FTI) threshold or strict criteria for contiguity was targeted. Ablation index was not available to the operators at the time of the procedure. In case of chest pain or intra-oesophageal T° rise >38.5 °C during posterior wall ablation, RF delivery was stopped prematurely. Assessment and management of first-pass isolation and acute PVR were identical as for the 'CLOSE' group. Cavotricuspid isthmus ablation was performed if appropriate.

Off-line analysis of criteria to assess integrity of the deployed radiofrequency circle

Each procedure was exported and analysed offline. For each RF tag within the circle, we determined time of application (s), median delivered power (W), impedance drop (Δ -Imp, Ω), average CF (CF, g), FTI (g), and AI (arbitrary unit, au). Custom-made software was used to automatically determine the path of encircling, the perimeter of the RF circle, ILD (centre-to-centre distance, mm), and ablation line contiguity index (ALCI). Ablation line contiguity index was developed as a criterion to algorithmically score the combination of contiguity and lesion depth and is defined as the ratio of the estimated width of a transmural lesion (determined by the AI) over its distance (ILD) to the closest transmural lesion.³

To determine the weakest link in a given RF circle, each circle was divided into 10 segments, each encompassing 3 to 5 RF tags.³ For each segment, we determined for each measure the minimal (Time_{min}, Power_{min},



$\Delta\text{-Imp}_{\min}$, $\text{CF}_{\min}$, $\text{FTI}_{\min}$, $\text{AI}_{\min}$, $\text{ALCI}_{\min}$) or maximal value ($\text{ILD}_{\max}$). In the CF group, 10 procedures were not available for off-line analysis.

We analysed CF variability using custom-made algorithm (Figure 2). For each application, the reconstructed instantaneous CF curve and histogram were analysed.^{5,6} Constant contact was defined as CF60% variability of absolute CF ≤ 10 g. Variable contact was defined as the CF60% variability of absolute CF > 10 g. Intermittent contact was classified when the percentage of absolute CF dropping below 3 g is more than 25% of the application.

Follow-up

Complications were reported on the CRF and collected during follow-up. After ablation, anticoagulation and previously failing anti-arrhythmic drug therapy (ADT) were continued. At 3 months, anticoagulation was continued according to stroke risk, whereas ADT was continued at the discretion of the treating physician. Clinical evaluation and ECG were performed at 1, 3, 6, and 12 months. Holter-ECG was performed at 3 months (24 h), 6 months (24 h), and 12 months (7 day) or in case of symptoms. Freedom from recurrence was defined as 1 year of absence from atrial fibrillation/atrial tachycardia/atrial flutter (AF/AT/ATF) > 30 s beyond a 3 months of blanking period.

Statistical analysis

Normality of data distribution was tested with the Shapiro–Wilk test. Continuous variables are expressed as mean $\pm$ standard deviation if normally distributed, medians with interquartile range if non-normally distributed and dichotomous variables as percentage. Respectively, variables were compared using the Student's *t*-test, Mann–Whitney *U* test, and Fisher's exact test. Linear mixed model tests were used to evaluate difference of ablation parameters between the CF-group vs. the CLOSE-group. The Kaplan–Meier estimate was used to compare freedom from

AF between both the groups during 12 months follow-up period. Multivariable logistic regression was used to assess the association between the different parameters and PVR. *P*-values < 0.05 were considered statistically significant. All statistical analyses were performed in SPSS Statistics 24 (IBM Corporation, Armonk, NY, USA).

Results

Patient characteristics

All patients had paroxysmal AF. There were no differences in baseline clinical characteristics between two groups (Table 1).

Radiofrequency characteristics and integrity of the radiofrequency circle

Two representative cases illustrating the difference in RF parameters and integrity of the deployed RF circles are given in Figure 3. Compared to the conventional CF-guided procedure (left panel), the 'CLOSE' procedure (right panel) is characterized by applications with shorter duration (ranging from 11 to 61 s), a higher rate of applications with high power (≥ 30 W), a higher rate of RF tags with a marked impedance drop ($\geq 7\Omega$), applications with lower FTI values (ranging from 184 to 875 gs) and a higher rate of applications with an AI of ≥ 550 . The range of obtained CF values (5 to 37 g) is not different from the conventional CF case. Finally, although the perimeter of the circles is comparable, the 'CLOSE'-guided circles are characterized by enclosing lesions all with an ILD ≤ 6 mm and an ALCI of $> 100\%$.

Weakest link analysis for all segments is summarized in Supplementary material online, Figure S1. Compared to the CF group,

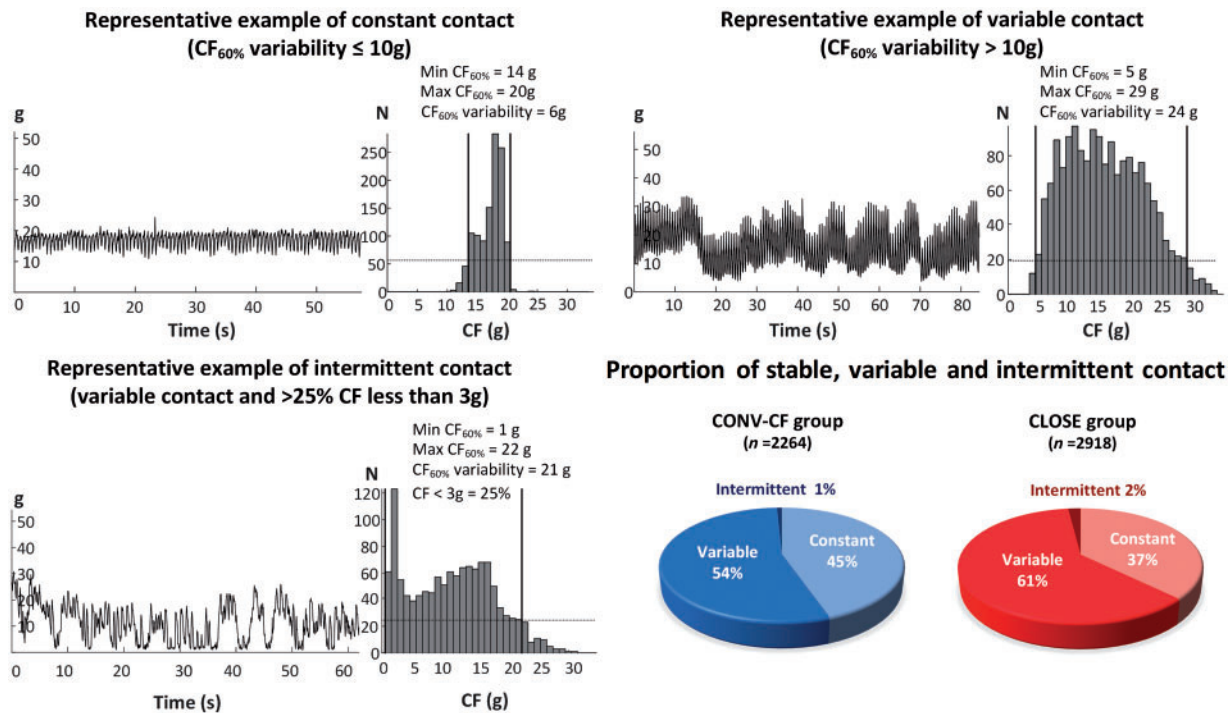


Figure 2 Representative examples of applications with constant contact, variable contact and intermittent contact. Right lower panel: the proportion of applications with intermittent contact is equally low in both groups. CF, contact force; CONV-CF, conventional contact force.

segments in the 'CLOSE' group are characterized by lower application time_{min} ($P < 0.001$), higher delivered power_{min} ($P < 0.001$), higher obtained Δ -Imp_{min} ($P < 0.001$), similar CF_{min} ($P = 0.401$), lower FTI_{min} ($P < 0.001$), higher AI_{min} ($P < 0.001$), lower ILD_{max} ($P = 0.032$), and higher ALCI_{min} ($P < 0.001$). Radiofrequency characteristics for the posterior and anterior segments separately are illustrated in the scatter plots.

Compared to the CF group, applications in the 'CLOSE' group were characterized by the same proportion of applications with intermittent contact (2% vs. 1%, $P = 0.67$) (Figure 2, right panels).

In the 'CLOSE' group, applications at the posterior wall lasted 23 ± 6 s overall. In 42 out of 100 circles (42%), there was at least one application with an AI target of 300. In those 'low AI-target' lesions, applications lasted for 13 ± 4 s only.

Procedural characteristics

Characteristics are given in Table 2. Pulmonary vein isolation was obtained in all circles. Cavotricuspid isthmus ablation was performed in 11 patients in the CF group and 4 in the 'CLOSE' group ($P = 0.091$). The use of general anaesthesia was comparable between groups. Compared to the CONV-CF group, the 'CLOSE' group was associated with a shorter procedure time (149 ± 33 min vs. 192 ± 42 min, $P < 0.0001$) (Figure 4). Perimeter of the right and left circles and number of applications per circle were not different. Compared to the CONV-CF group, the 'CLOSE' group was associated with shorter RF time per circle (17 ± 3 min vs. 29 ± 8 min, $P < 0.0001$ for the right circles, 19 ± 4 min vs. 26 ± 6 min, $P < 0.001$ for the left circles).

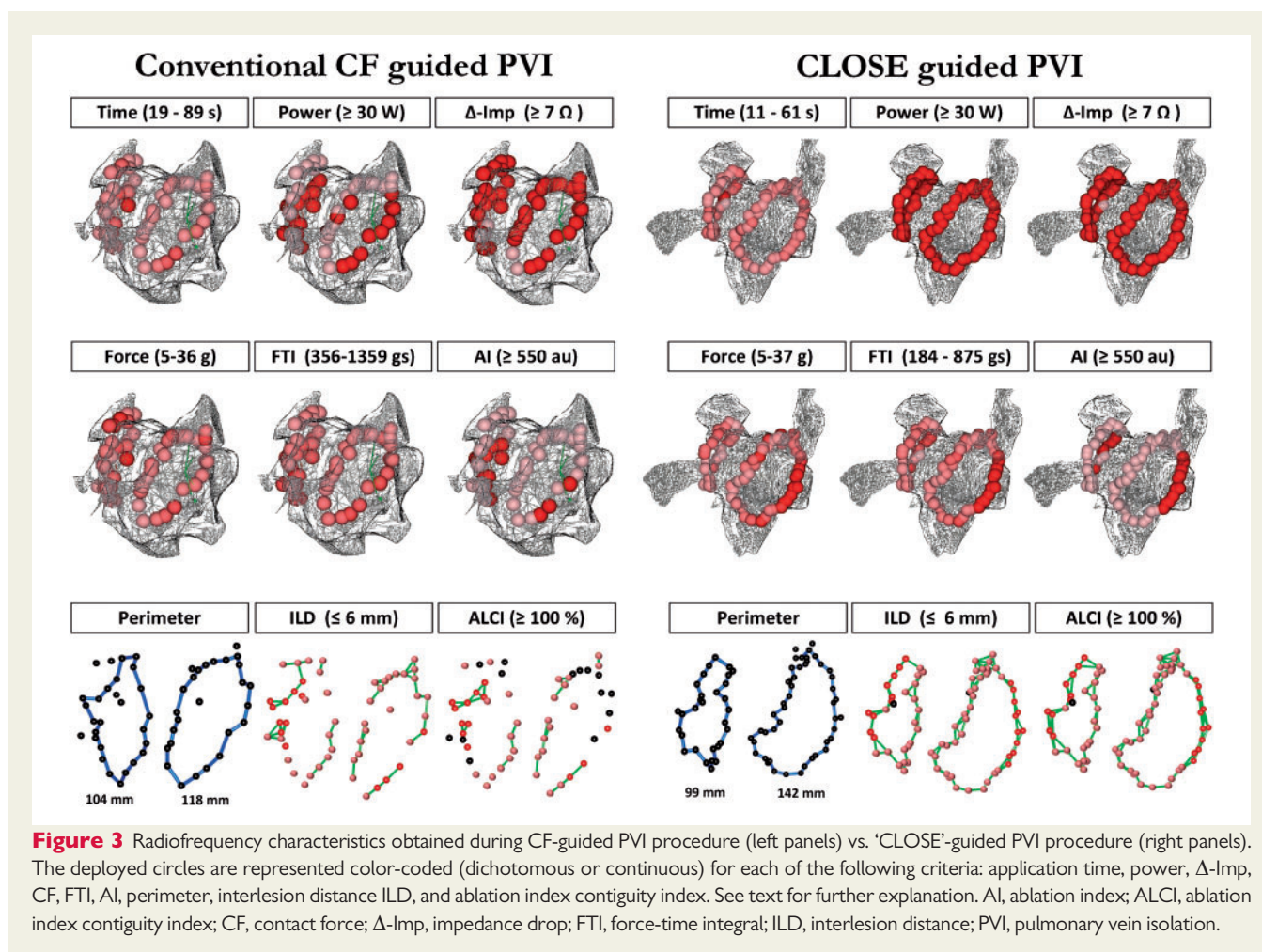
Table 1 Patient characteristics

	Conventional CF group (n = 50)	CLOSE group (n = 50)	P-value
Paroxysmal AF	50 (100%)	50 (100%)	>0.99
Age (years)	62 ± 10	61 ± 11	0.466
Male, n	31 (62%)	36 (72%)	0.395
BMI (kg/m ²)	26.6 ± 4.3	27.0 ± 3.7	0.678
Arterial hypertension, n	20 (40%)	15 (30%)	0.402
Structural heart disease, n	7 (14%)	3 (6%)	0.318
Diabetes, n	0 (0%)	2 (4%)	0.495
CHA ₂ DS ₂ -VASc score ≥ 2, n	25 (50%)	16 (32%)	0.068
Left atrial diameter (mm)	40.4 ± 4.4	39.9 ± 3.9	0.597
Electrical cardioversion, n	7 (14%)	3 (22%)	0.436
1st AF episode to PVI (months)	24 [12–72]	24 [12–84]	0.623
Resistance to ADT, n	47 (94%)	45 (90%)	0.715

ADT, anti-arrhythmic drug therapy; AF, atrial fibrillation; CF, contact force; PVI, pulmonary vein isolation.

First-pass and waiting time/adenosine-proof isolation

Results are summarized in Figure 4. A limited number of patients did not receive adenosine because of a history of asthma or COPD (five in the CONV-CF group vs. three in the 'CLOSE'-group). Compared



to the CONV-CF group, the 'CLOSE' group was associated with higher incidence of first-pass isolation (98% vs. 54%, $P < 0.0001$) and waiting time/adenosine-proof isolation (97% vs. 82%, $P < 0.001$).

One year of freedom from atrial fibrillation/atrial tachycardia/atrial flutter

The Kaplan–Meier analysis was plotted in Figure 5. At the end of the blanking period, 23 patients continued ADT in the CONV-CF group (46%) vs. 18 patients in the 'CLOSE' group (36%).

Overall, single-procedure 12 month freedom from AF/AT/AFL was 94% in the CLOSE compared to 80% in the CONV-CF group ($P < 0.05$). Both in patients with and without ADT, 'CLOSE' was associated with a higher freedom from AF/AT/AFL (Figure 5, right panels).

Safety of 'CLOSE'- guided vs. conventional contact force-guided pulmonary vein isolation

No symptomatic complications were observed in the 'CLOSE' group. In the conventional CF group, we observed one tamponade during transseptal puncture requiring pericardiocentesis and prolonged hospitalization. Finally, there was no significant difference in hospitalization length.

Predictors of atrial fibrillation recurrence and mechanism of reconnections at repeat

Compared to patients with successful 1 year outcome ($n = 87$), patients with recurrence ($n = 13$) had similar baseline clinical characteristics. Use of 'CLOSE' protocol was an independent predictor for maintenance of sinus rhythm (54% vs. 23%; OR 3.917; 95% CI 1.008–15.220).

In the CONV-CF group (seven patients undergoing repeat), there were 23 sites of reconnection in 18 veins, in the CLOSE group (three patients undergoing repeat), we observed seven sites of reconnection in seven veins. We observed no difference in spatial distribution of reconnection sites in between both groups (43% posterior in the CLOSE group vs. 48% posterior in the CF group) (Figure 6). In both groups, the majority of reconnections were associated with either ILD > 6 mm and/or AI $< 400/550$ (7 of 7, 100% in 'CLOSE' vs. 19 of 23, 83% in 'CONV-CF', $P = 0.99$).

Discussion

Main findings

This study shows that, compared to conventional CF-guided ablation, an ablation strategy aiming to enclose the PVs with contiguous and

Table 2 Procedure characteristics

	Conventional CF group (n = 50)	CLOSE group (n = 50)	P-value
PV isolation, n	50 (100%)	50 (100%)	>0.99
CTI ablation, n	11 (22%)	4 (8%)	0.091
General anesthesia, n	18 (36%)	25 (50%)	0.225
Procedure time (min)	192 ± 42	149 ± 33	<0.001
Total RF time (min)	56 ± 11	36 ± 7	<0.001
Right circles			
Perimeter (mm)	138 ± 24	129 ± 19	0.068
RF tags, n	41 ± 10	38 ± 8	0.077
RF time (min)	29 ± 8	17 ± 3	<0.001
First-pass isolation, n	25 (50%)	50 (100%)	<0.001
Waiting/adeno. proof isolation, n	41 (82%)	49 (98%)	0.008
Left circles			
Perimeter (mm)	123 ± 26	122 ± 27	0.985
RF tags, n	37 ± 10	40 ± 10	0.068
RF time (min)	26 ± 6	19 ± 4	<0.001
First-pass isolation, n	29 (58%)	48 (96%)	<0.001
Waiting/adeno. proof isolation, n	41 (82%)	48 (96%)	0.025

CF, contact force; CTI, cavotricuspid isthmus; PV, pulmonary vein; RF, radiofrequency.

optimized RF lesions results in a higher rate of adenosine-proof isolation and higher single-procedure arrhythmia-free survival. A single-procedure freedom of 94% at 1 year is obtained without compromising safety and with shorter procedure and RF times. Whereas in the absence of randomization, this study should be interpreted as hypothesis generating only, our data might provide the necessary base to setup a randomized control trial.

Single-procedure freedom of atrial fibrillation after point-by-point radio-frequency pulmonary vein isolation

The recent FIRE and ICE study confirmed a 64% single-procedure freedom from AF after point-by-point RF ablation.^{7,8} Since the introduction of CF-sensing technology in ablation catheters, several studies suggested enhanced 1 year of freedom from AF.^{9–12} In a two-centre prospective parallel cohort study, CF-guidance was associated with better 1 year of arrhythmia-free survival (88% vs. 66%).⁹ In a single centre controlled study including 60 paroxysmal AF patients, CF-guidance was associated with higher 1 year of freedom from AF (90% vs. 67%).¹⁰ Although randomized controlled studies failed to demonstrate a beneficial effect of CF-guidance on AF free survival at 1 year,^{13,14} our data confirm the ≈80% single-procedure freedom of AF after CF-guided PVI (historical control group).

More importantly, the present data suggest that further optimization of CF-guided ablation by adding objective criteria (the 'CLOSE' criteria of ILD and AI) further improves acute durable isolation rates and 1 year single-procedure outcome. Several observations speak for

the reliability of the currently reported >90% single-procedure freedom of AF after 'CLOSE'-guided PVI. First, this high success rate is in line with the hypothesis that avoiding weak links within the deployed RF circle is the key to durable PVI and clinical success.³ Second, also other research groups, respecting a contiguous encircling, reported a >90% single-procedure arrhythmia-free survival in CF-guided PVI. A recent sub-analysis of SMART-AF in 40 paroxysmal AF patients showed 93% freedom from arrhythmia in those patients with short ILD.¹⁵ Likewise, Itoh et al.¹⁶ reported single-procedure freedom of 96% at 1 year after a procedure in which CF-guided PVI was performed using catheter tip dragging in steps of ≈2 mm. Finally, the currently reported high single-procedure freedom after PVI for paroxysmal AF is in line with what one can expect after durable PVI. On the one hand, it is known that non-PV triggers account for <10% of paroxysmal AF,¹⁷ on the other hand a strategy of aiming for durable PVI with repeat ablations is known to cumulate in a >90% freedom of AF.¹⁸

Conducting a randomized controlled trial confirming the superiority of the 'CLOSE'-protocol will be challenging. In a prospective control group claiming 'conventional CF' only, it seems impossible to avoid the bias of previous experience with AI and ILD. Also other study designs using other control groups will face challenges. Although no randomized trial showed incremental benefit of CF-sensing catheters, it seems unethical to design a control group using catheters without CF sensing. Finally, a control group using catheters with other energy sources will be prone to the obvious criticism of bias introduced by operator experience or preference. Therefore, we believe that the current database of 100 consecutive cases offers a unique cross-sectional view allowing the comparison of both strategies.

What could explain the higher success rate of the 'CLOSE' protocol?

The improved procedural and 1 year outcome in 'CLOSE' are in line with the weakest link hypothesis.³ Whereas in the conventional CF group there were no objective ranges (not even for CF), the RF circle in the 'CLOSE' protocol was characterized by lesions all with an ILD ≤ 6 mm and an AI ≥ 400 (300) at posterior and 550 at anterior wall. Those target values, resulting in an ablation line contiguity index of 100%, were found to be predictive for durable isolation in CF-guided PVI.³ The observation that the majority of reconnections in both groups were due to either ILD > 6 mm and/or AI < 400/550 also supports this hypothesis.

Nevertheless, improved outcome might be (partly) due other factors beyond ILD and AI as well. First, one might speculate that the improvement in the 'CLOSE' protocol is mediated by the use of higher power (35 W at the posterior wall instead of 25 W) and shorter application times to obtain AI targets. Although we did not observe any major shift in the spatial distribution of gaps at repeat in between both groups and although there exists preclinical evidence that lesion size is similar at a certain AI value (despite differences in power, force, and time of application to reach that value),¹⁹ no clinical study compared quality of lesions being delivered with different settings. Second, one might hypothesize that improved outcome is due to reduced CF variability during RF delivery in the 'CLOSE' protocol. Stability of lesion delivery, as shown by Shah et al.⁵ is a critical

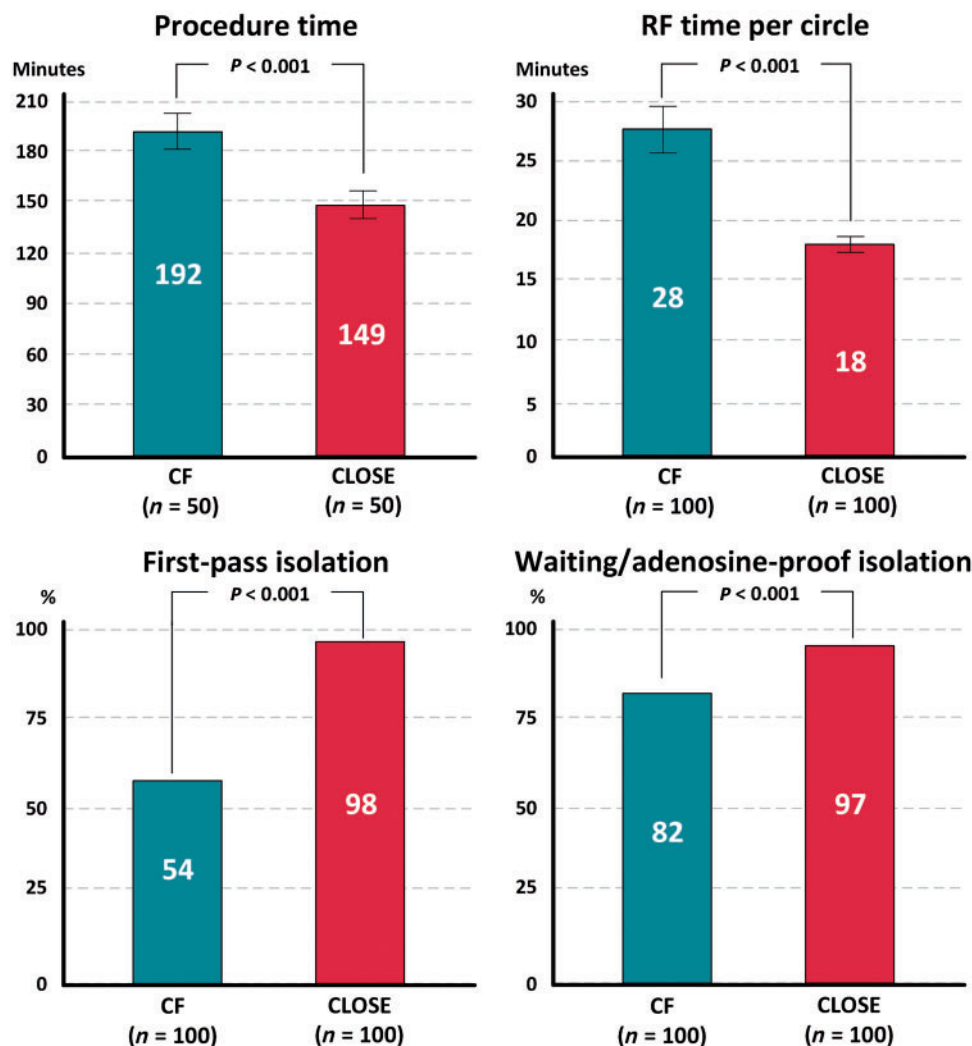


Figure 4 Upper panels: procedure time and radiofrequency time per circle. Lower panels: incidence of first-pass isolation and waiting time/adenosine-proof isolation in both groups. See text for further explanation. CF, contact force; RF, radiofrequency.

determinant of lesion size,⁶ and one might expect less CF variability during shorter applications. Of interest however, we did not observe any difference in CF variability in between both groups. In fact, the proportion of applications with intermittent contact was equally low in both groups, due to the use of combined stability and force criteria in VisiTag module. Finally, improved outcome could be (partly) due to a learning curve promoted by the experience of technology and algorithm application. In the absence of a randomized trial, this hypothesis can never be refuted.

Safety precautions during 'CLOSE'-guided ablation at the posterior wall

The present patient cohort is too small to draw any definitive conclusions regarding safety in 'CLOSE'-guided PVI. In a recent study, CF catheters were associated with a high incidence of atrial-oesophageal fistula.²⁰ Therefore, one could argue that a CF-guided strategy with higher power ablation on the posterior wall aiming at fixed AI target

values might be imprudent.²¹ So far, however, we did not detect any clinically relevant oesophageal injury using the 'CLOSE' protocol.⁴ This might be due to the following reasons: (i) in the 'CLOSE' protocol, we strive for high stability rather than high CF during RF delivery. Avoiding high force might avoid pushing the posterior wall towards the oesophagus; (ii) In our approach, RF is delivered at the left atrial-PV junction rather than at the thin free wall of the posterior left atrium; (iii) AI target was reduced to 300 at the posterior wall in the presence of pain perception or temperature rise (general anaesthesia). This lower target was also maintained for the rest of the posterior wall lesions of that circle. As a result, 42% of 'CLOSE' circles were characterized by one or more lesions with an AI < 400; (iv) Finally, due to the use of relatively higher power (35 W), the 'CLOSE' protocol is associated with short applications at the posterior wall (23 ± 6 s), even as short as 13 ± 4 s in case of possible signs of oesophageal injury. Short applications might be associated with a lower risk of deep penetration of conductive heating to adjacent structures such as the oesophagus.

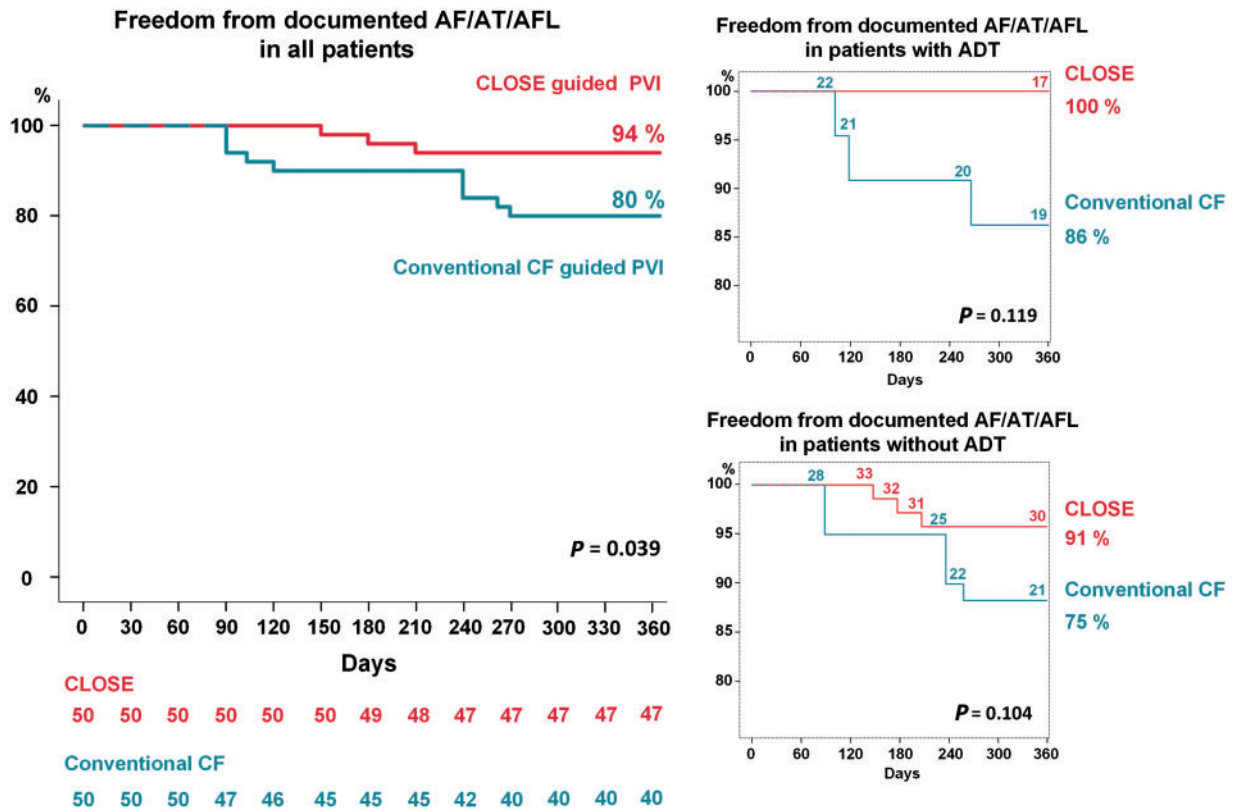


Figure 5 Left panel: The Kaplan–Meier curves depicting time to first recurrence of atrial fibrillation (AF), atrial flutter (AFL), or atrial tachycardia (AT) (including a 3 months of blanking period) in patients undergoing ‘CLOSÉ’-guided ablation and conventional CF-guided ablation. Right panels: Arrhythmia-free survival for patients who continued anti-arrhythmic drug therapy (ADT) at the end of the blanking period (upper panel) and patients without ADT (lower panel). CF, contact force; PVI, pulmonary vein isolation.

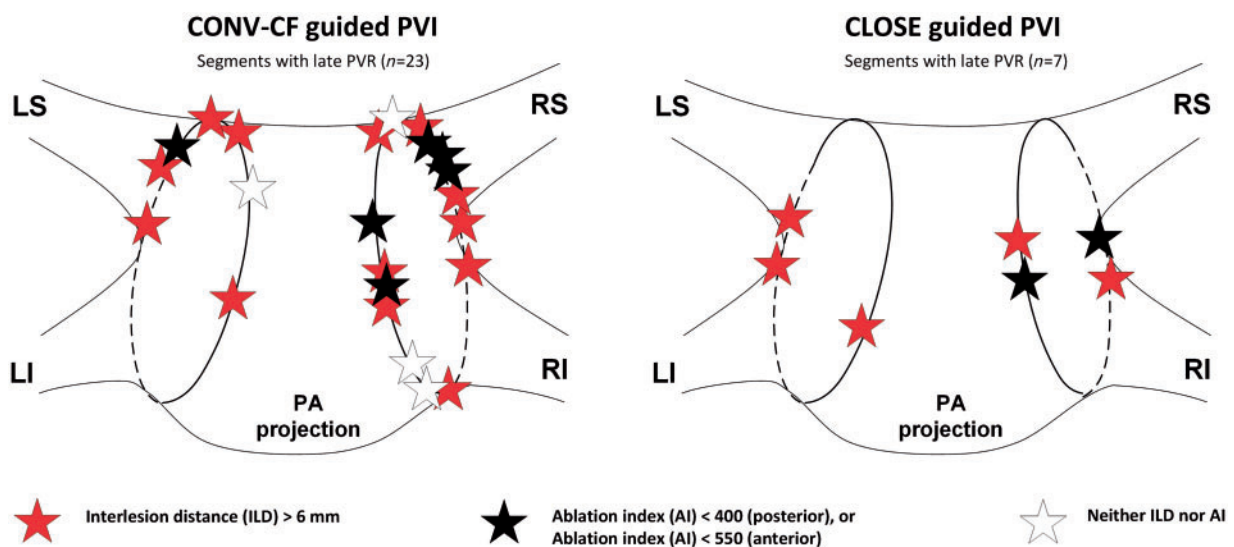


Figure 6 Spatial distribution of the sites of reconnection in both treatment groups and their putative mechanism for reconnection. See text for explanation. CONV-CF, conventional contact force; LI, left inferior; LS, left superior; PA, postero-anterior; PVI, pulmonary vein isolation; PVR, pulmonary vein reconnection; RI, right inferior; RS, right superior.

Of course, safety (and its putative determinants) needs to be investigated in properly powered clinical trials or in studies investigating oesophageal thermal asymptomatic lesions using oesophageal endoscopy.

Towards a short and standardized pulmonary vein isolation procedure

In the present study, we observed that 'CLOSE'-guided PVI is associated with a significant 30% reduction in procedure time and nearly 50% reduction in RF time. A shorter procedure time in the 'CLOSE' protocol is explained by the absence of PV reconnections (no time spent chasing gaps at the junction) and a reduction in RF time. Shorter RF time is explained by the markedly shorter applications (use of AI allowing delivery of higher power) and a trend towards less applications. Although the 192 ± 42 min procedure time and 56 ± 11 min total RF time, as observed in the 'CONV-CF' group, are in line with procedure time and RF time previously reported in CF-guided procedures (223 ± 86 min and 61 ± 36 min in the SMART-AF study¹¹ and 194 ± 23 min and 41 ± 7 min in the multicentre UK study¹³) we can never exclude the contribution of a learning curve.

Finally, we believe that the 'CLOSE' protocol offers the base for a straightforward and standardized PVI procedure. Rather than isolation of the veins, the endpoint for the procedure is a perfect circle leading invariably to PVI. Introduction of objective and verifiable criteria (AI and ILD) should lead to reproducible outcome across centers and operators.

Limitations

Although 100 patients were enrolled in this pragmatic parallel cohort study, several limitations need to be underlined: our study is monocentric, non-randomized, and small-sized. Despite a 24 h-Holter ECG at 3, 6 months, and 7 days of Holter-ECG at 12 months and the symptomatic nature of PAF, the discontinuous nature of monitoring can underestimate AF recurrence. Finally, ADT was not systematically withdrawn after the blanking period and may have improved clinical success in both groups. The limitations of discontinuous monitoring and single-centre nature of the study are being tackled by two ongoing clinical studies (a study monitoring arrhythmia-free survival using subcutaneous loop recorders and a multicentre clinical trial in >300 patients). Finally, the present comparison is too small to draw any definitive conclusions regarding safety.

Conclusions

In the present study, we evaluated the feasibility of an ablation protocol aiming to enclose the pulmonary veins with contiguous and optimized RF lesions. Compared to conventional CF-guided PVI, we observed a shorter procedure time, significant improvement in acute PVI durability and a higher single-procedure arrhythmia-free survival at 1 year.

Supplementary material

Supplementary material is available at *Europace* online.

Conflict of interest: M.D. received minor compensations from Biosense Webster Inc., Abbott, and Biotronik as an invited speaker. M.E.H. receives consultancy fees from Biosense Webster Inc.

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